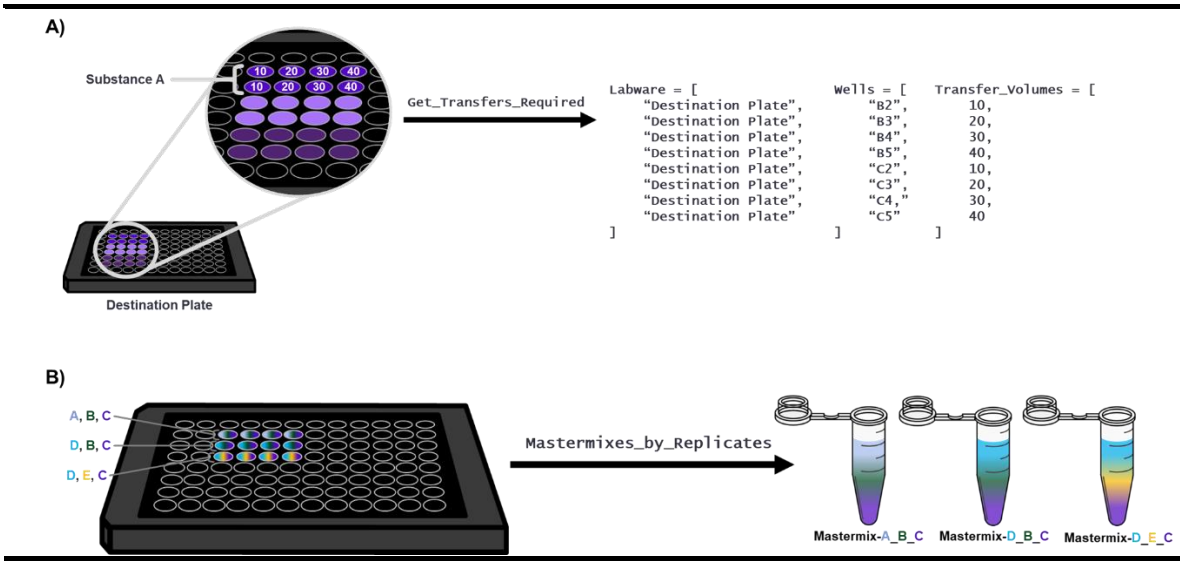
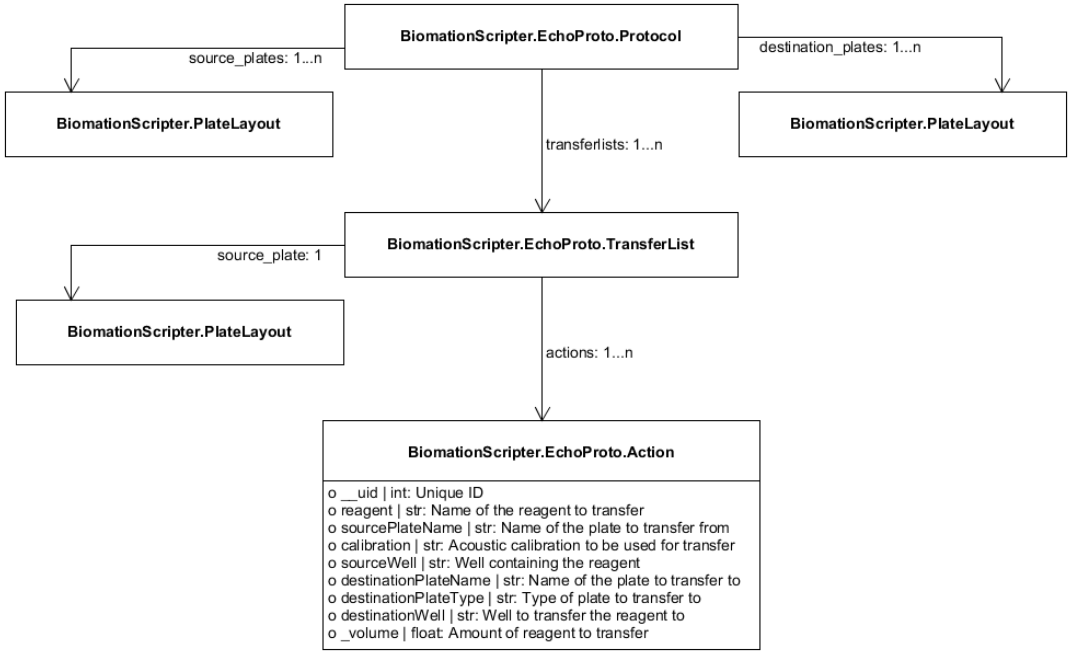
BiomationScripter: A Python Library for Developing Synthetic Biology Automation Protocols

-- Supplementary Figures



Supplementary Figure 1. Generic BiomationScripter Helper Functions: Graphical representations for a selection of generic BMS helper functions. **A)** Information regarding transfer actions for a specific liquid into a destination labware can be automatically generated using `Get_Transfers_Required`. The destination labware, well, and transfer volume are generated. This information can be directly used by other BMS functions. B**)** Mastermixes can be automatically generated using `Mastermixes_By_Replicates`. Within a list of destination labware, wells with identical proportions of the same source materials are grouped, and mastermixes for those wells are generated and added to a `Labware_Layout` object. The destination labware is then updated to indicate which wells should be supplied by which mastermixes.



Supplementary Figure 2. EchoProto Architecture: Architecture of the EchoProto module. All information for an EchoProto protocol is captured by the `Protocol` class. Aside from metadata such as the protocol name, the available source plates are stored using the `Labware_Layout` class from BMS. Destination plates containing the desired content after the protocol has completed are also captured using the `Labware_Layout` class. Information regarding transfer events from each source plate are captured using the `TransferList` class, and information about the transfers themselves is captured using the `Action` class.